Exam 2 Practice Answers

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Question 1

1. Return to the email dataset from Lecture 7. This time, let's examine the variable "line_breaks", including the following steps:

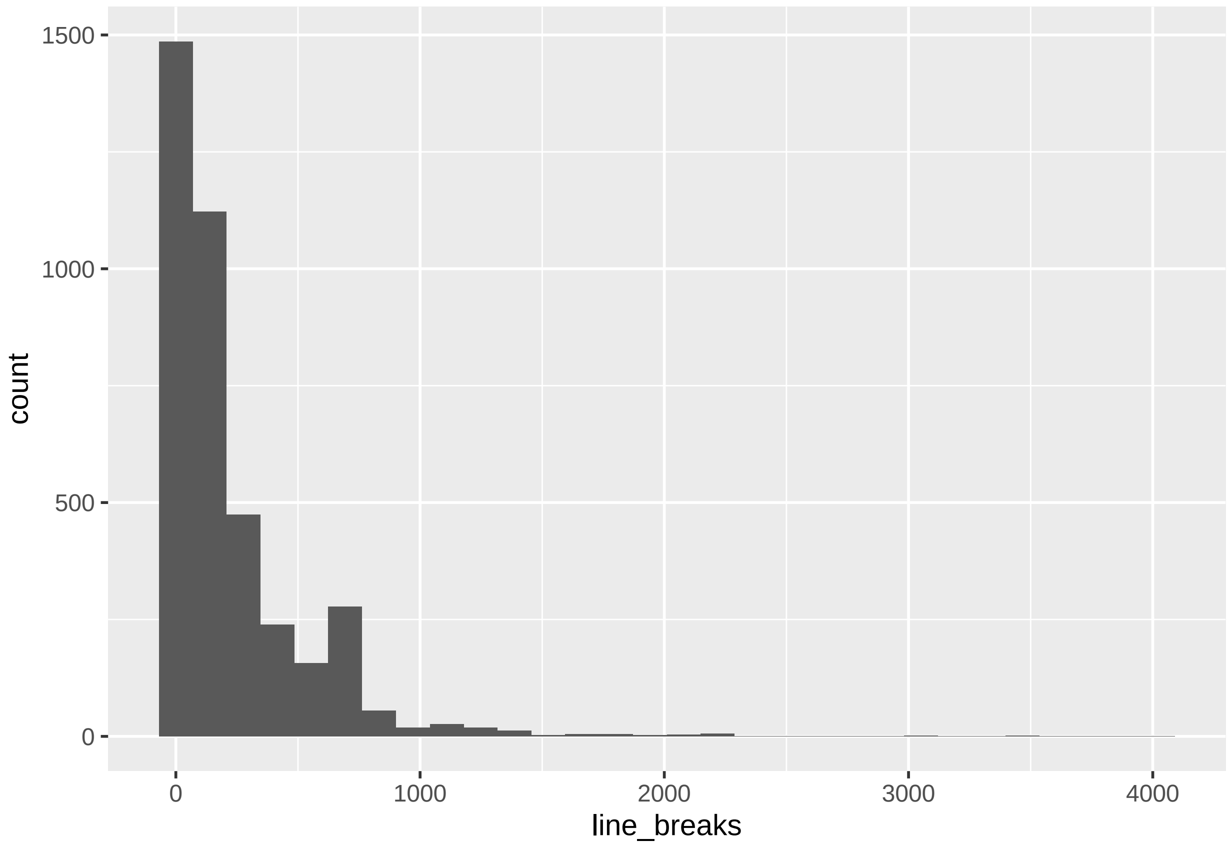
a. Plot the variable using an appropriate R function.
b. Identify the sample mean, sample size, and sample standard deviation.
c. Obtain a 95% two-sided confidence interval for this variable.
d. Perform a hypothesis test at confidence level alpha=0.05 to investigate whether or not the population mean for line breaks is 250. List both the test statistic and its resulting p-value.
e. Do you reject or fail to reject the null hypothesis?
f. What can you conclude about the mean number of line breaks given the above hypothesis test?

1A

```
email <- read.csv("https://raw.githubusercontent.com/roualdes/data/master/email.csv")

suppressWarnings(library(ggplot2))
ggplot(data=email, aes(line_breaks)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
summary(email$line_breaks)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1.0    34.0   119.0   230.7   298.0  4022.0
```

1B

```
xbar = mean(email$line_breaks,na.rm=T) # sample mean
n = length(email$line_breaks) - sum(is.na(email$line_breaks))
s = sd(email$line_breaks,na.rm=T)
```

1C

```
# 95% CI 2-sided
t.star = qt(c(0.025,0.975),df=n-1)
se = s/sqrt(n)
CI = xbar + t.star*se
CI
```

```
## [1] 220.6611 240.6560
```

1D

```
# H0: mu = 250
# H1: mu != 250
alpha = 0.05
mu = 250
t.stat = (xbar - mu)/(se)
t.stat
```

```
## [1] -3.793
```

```
p.val = 2*(1-pt(abs(t.stat),df=n-1)) # 2-sided p-value
p.val
```

```
## [1] 0.0001510817
```

```
p.val < alpha
```

```
## [1] TRUE
```

1E

Do we reject H0? YES, we reject H0 because our p.value is less than the confidence level alpha.

1F

We have sufficient evidence to claim, at confidence level alpha = 0.05, that the mean number of line breaks in the population is NOT 250.

Question 2

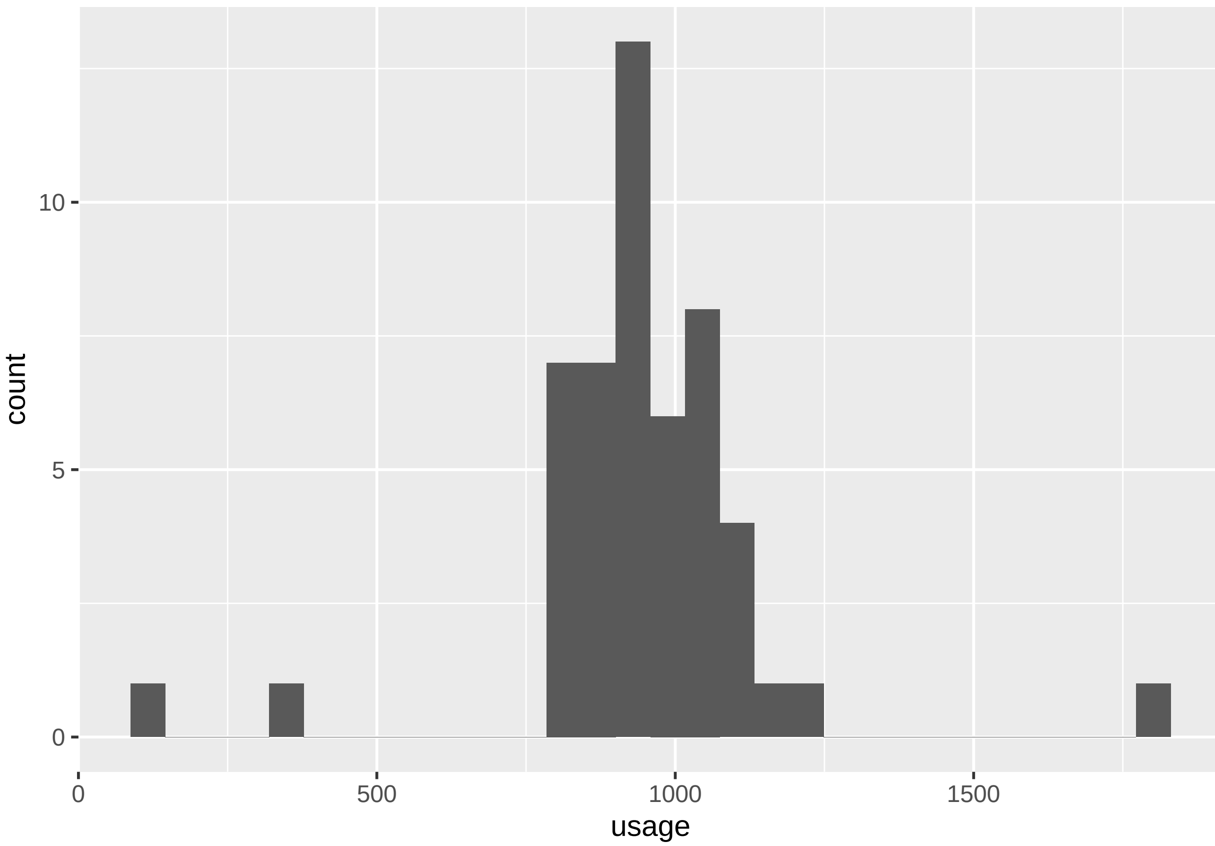
2. The average monthly energy use for a typical household was previously determined to be 909 kWh, with a known population standard deviation of 105 kWh. Say a survey is done to test whether the average energy use is now higher than this. Download the "energy.csv" file to view the results of the survey, which list the energy use in kWh of several homes.

a. Plot the variable using an appropriate R function. Do you notice any unusual values or outliers in the data? Why might such a value exist for this data set?
b. Identify the sample mean, sample size, and sample standard deviation for this variable.
c. Use the population standard deviation to find a 95% confidence interval for mean energy usage.
d. Perform an appropriate hypothesis test at confidence level alpha=0.05 to investigate the above suspicion for this data. List both the test statistic and its resulting p-value.
e. Do you reject or fail to reject the null hypothesis?
f. What if you instead decide to use the sample standard deviation to perform a t-test? Find both the confidence interval and the hypothesis test for this situation.
g. Do your results differ from those of part e? Why or why not?

2A

```r
# Load in the data (this will vary in location depending on where your file is,
# and what your current working directory is)
energy = read.csv("energy.csv")
ggplot(data=energy, aes(usage)) +geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



There are a few outliers in this data set. The high-value outlier could be for a very large house or mansion, while the low-value outliers may be temporarily vacated or vacation homes (among other possibilities).

2B

```r
n = length(energy$usage)
xbar = mean(energy$usage)
s = sd(energy$usage) # NOTE: Not used until 2F
```

2C

```r
sigma = 105
alpha = 0.05
# ONE-SIDED UPPER CONFIDENCE INTERVAL (See Lec06_Examples.pdf)
z.star = qnorm(1-alpha)
se = sigma/sqrt(n)

CI = c(xbar - z.star*se, Inf)
CI
```

```
## [1] 914.7259      Inf
```

2D

```r
# H0: mu = 909
# H1: mu > 909

z.stat = (xbar - 909) / (sigma / sqrt(n))
z.stat
```

```
## [1] 2.030453
```

```r
p.val = 1-pnorm(z.stat) # One-sided p-value
p.val
```

```
## [1] 0.02115523
```

```r
p.val < alpha
```

```
## [1] TRUE
```

2E

Do we reject H0? YES, we reject H0 since our p-value is less than our confidence level alpha. We have sufficient evidence (at alpha = 0.05) to suggest that mean household monthly energy use is higher than 909 kWh.

2F

```r
# CI 95%
sigma = 105
alpha = 0.05
t.star = qt(1-alpha,df=n-1)
se = s/sqrt(n)
CI = c(xbar - t.star*se, Inf)
CI
```

```
## [1] 888.3464      Inf
```

```
# Hypothesis Test
# H0: mu = 909
# H1: mu > 909
mu = 909

t.stat = (xbar-mu)/(se)
t.stat
```

```
## [1] 0.9949787
```

```
p.val = 1-pt(t.stat,df=n-1)
p.val
```

```
## [1] 0.1623174
```

```
p.val < alpha
```

```
## [1] FALSE
```

2G

Do we reject H0? NO, we now fail to reject H0 since our p-value is now greater than the confidence level alpha. We have insufficient evidence (at alpha = 0.05) to suggest that the mean household monthly energy use is greater than 909kWh. This is because of the much larger sample standard deviation, which increases the size of the confidence interval and makes our sample mean not as unlikely to occur.

Question 3

3. A medical study recorded the cholesterol levels of 35 patients in both 2010 and 2020, and wishes to determine whether there is a difference in cholesterol levels for each patient. Download the data set "cholesterol.csv" for this question.

a. Would this data call for a paired t-test or a two-sample t-test?
b. Plot the variable using an appropriate R function.
c. Identify all relevant quantities (such as the estimator, sample sizes, and sample standard deviations) for this variable.
d. Obtain a 98% two-sided confidence interval for this variable.
e. Perform an appropriate hypothesis test at confidence level alpha=0.02. List both the test statistic and its resulting p-value.
f. Do you reject or fail to reject the null hypothesis?
g. What can you conclude about the patients' cholesterol levels based on your results in part f?

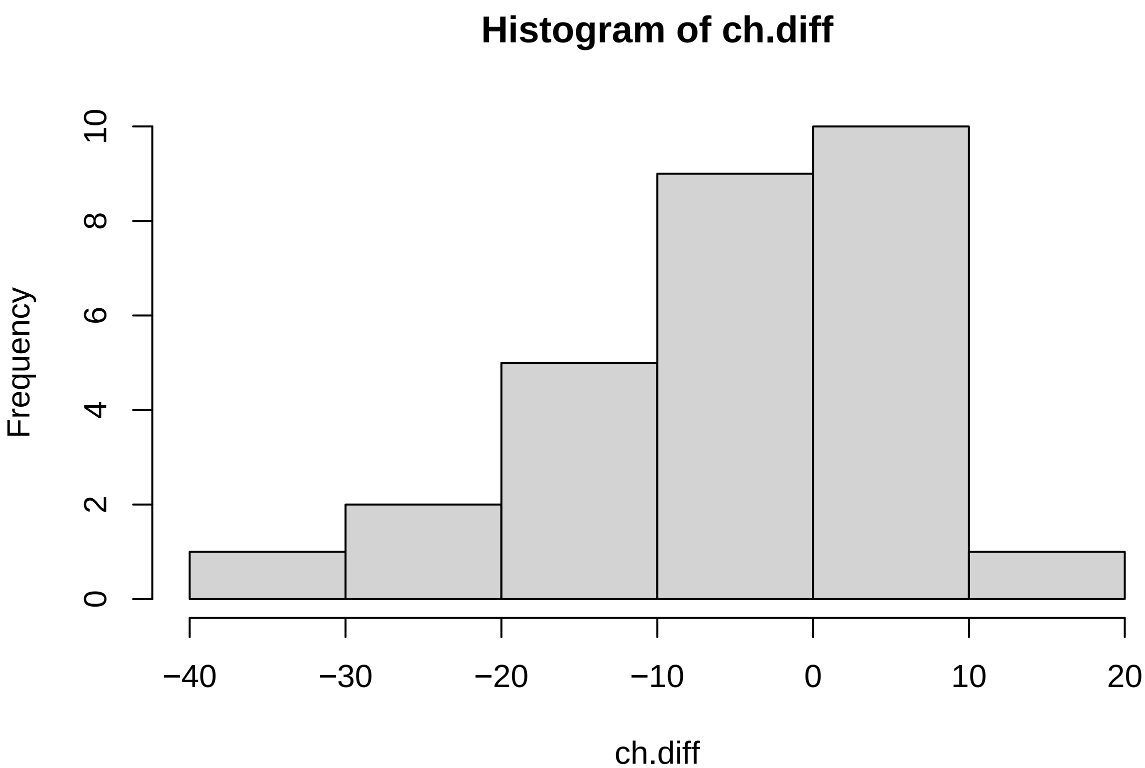
3A

```
# Load in the data (this will vary in location depending on where your file is,
# and what your current working directory is)
ch = read.csv("cholesterol.csv")
```

Is this data paired as opposed to two-sample? YES, it is paired. The observations are done for the SAME 35 patients, so the data is NOT independent–one patient's 2010 levels connect clearly to their 2020 levels.

3B

```
ch.diff = ch$after-ch$before # Obtain an object for the difference
hist(ch.diff)
```



3C

```
xbar.diff = mean(ch.diff) # This is the estimator
s = sd(ch.diff)
n = length(ch.diff)
```

3D

```
t.star = qt(c(0.01,0.99),df=n-1)
se = s/sqrt(n) # Take sample error for the difference
# INSTEAD of combining the two groups' separate sample errors.

CI = xbar.diff + t.star*se
CI
```

```
## [1] -10.1744455   0.2088336
```

3E

```
#H0: mu.diff = 0
#H1: mu.diff != 0
alpha = 0.02

t.stat = (xbar.diff - 0)/se
t.stat
```

```
## [1] -2.373197
```

```
p.val = 2*(1-pt(abs(t.stat),df=n-1))
p.val
```

```
## [1] 0.02501225
```

```
p.val < alpha
```

```
## [1] FALSE
```

3F

Do we reject H0? YES, we reject H0, since our p-value is less than our confidence level alpha. We conclude that we have sufficient evidence (at alpha = 0.02) to suggest that there is a difference between the patients' 2010 and 2020 cholesterol levels (though we can't say what kind of difference with a two-sided test).

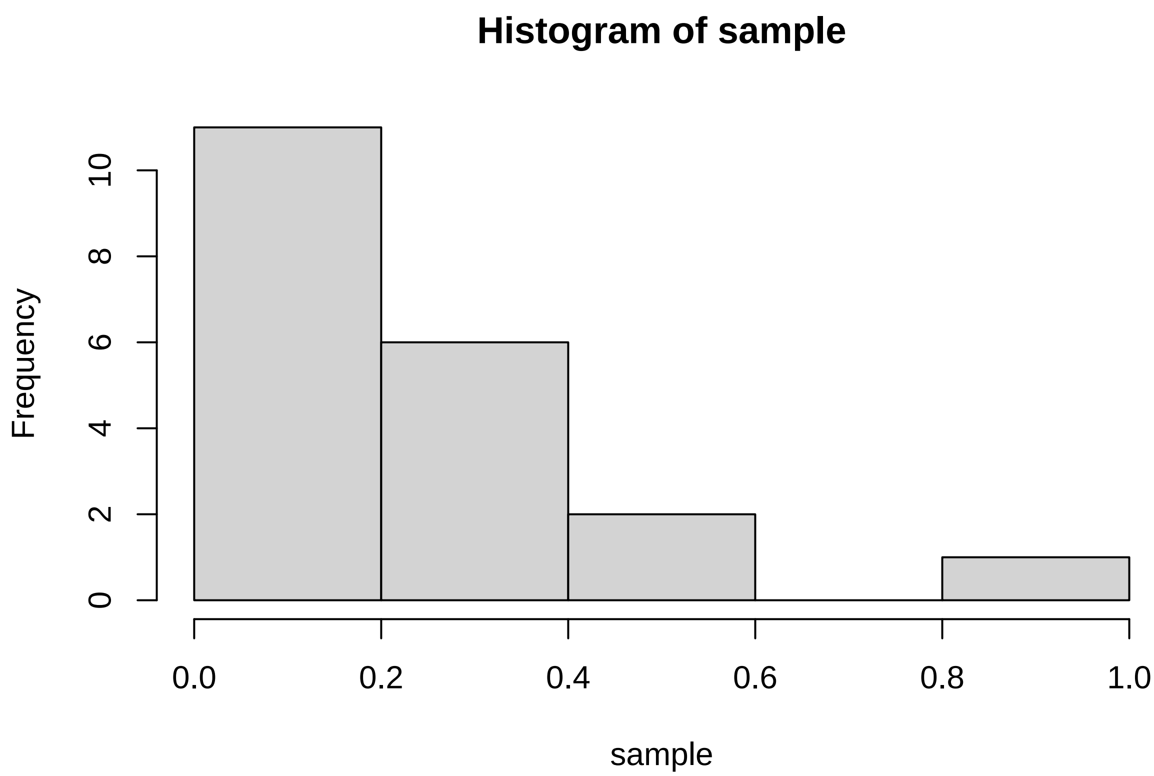
Q4

4. In RStudio, generate a data set of size n=20 from the Exponential distribution, with X ~ Exp(rate=5).

a. Create a histogram for this sample. What skew, if any, does the data have?
b. Calculate R=500 sample statistics Xbar for r=1,...,R.
c. Create a histogram for your sample statistics. How does the sampling distribution for the estimator Xbar compare to that of the initial sample?
d. Repeat the above steps for a sample (and multiple samples) of size n=200. How do the population and sampling distributions change?

4A

```
# You can use ?rexp to double check how to generate random data
set.seed(0)
sample = rexp(n=20,rate=5)
hist(sample)
```



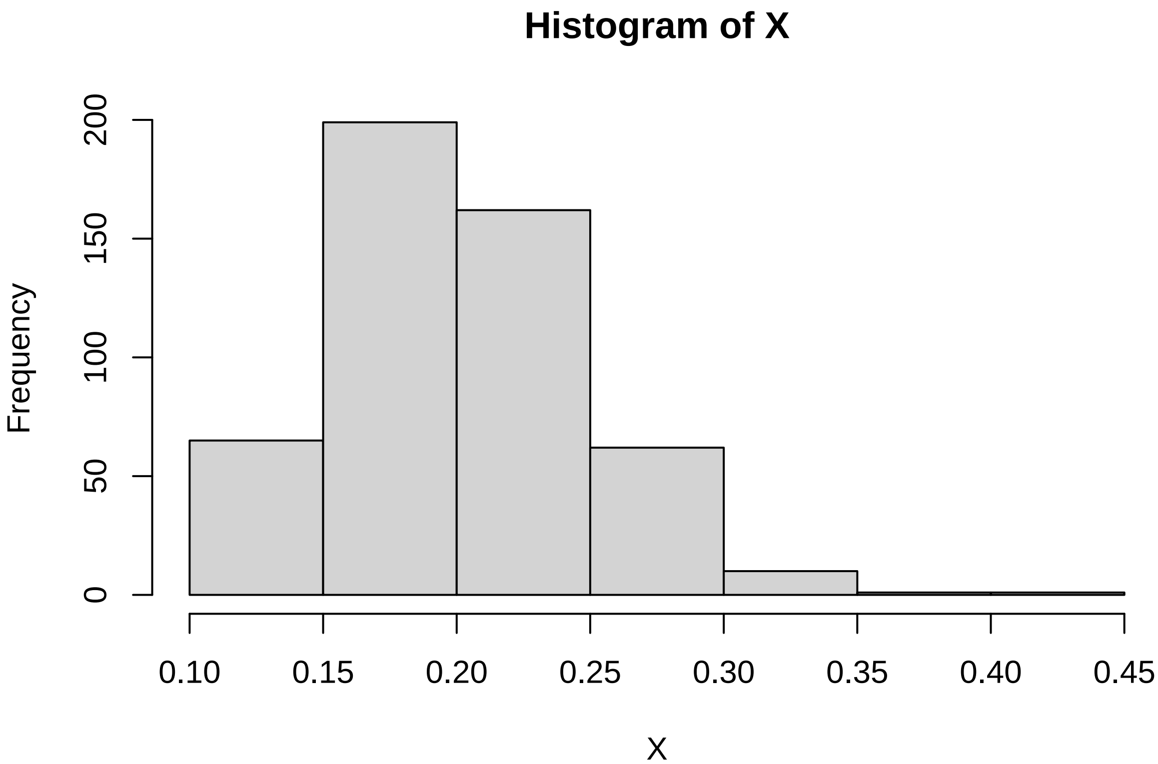
The data appear to be right-skewed for this sample.

4B

```
R = 500 # For 500 samples
X = rep(NA,R) # This will store our Xbar values
for(i in 1:R)
{
  sample = rexp(n=20,rate=5) # Generate a sample...
  X[i] = mean(sample) # ...and record its mean as the ith entry of X.
}
```

4C

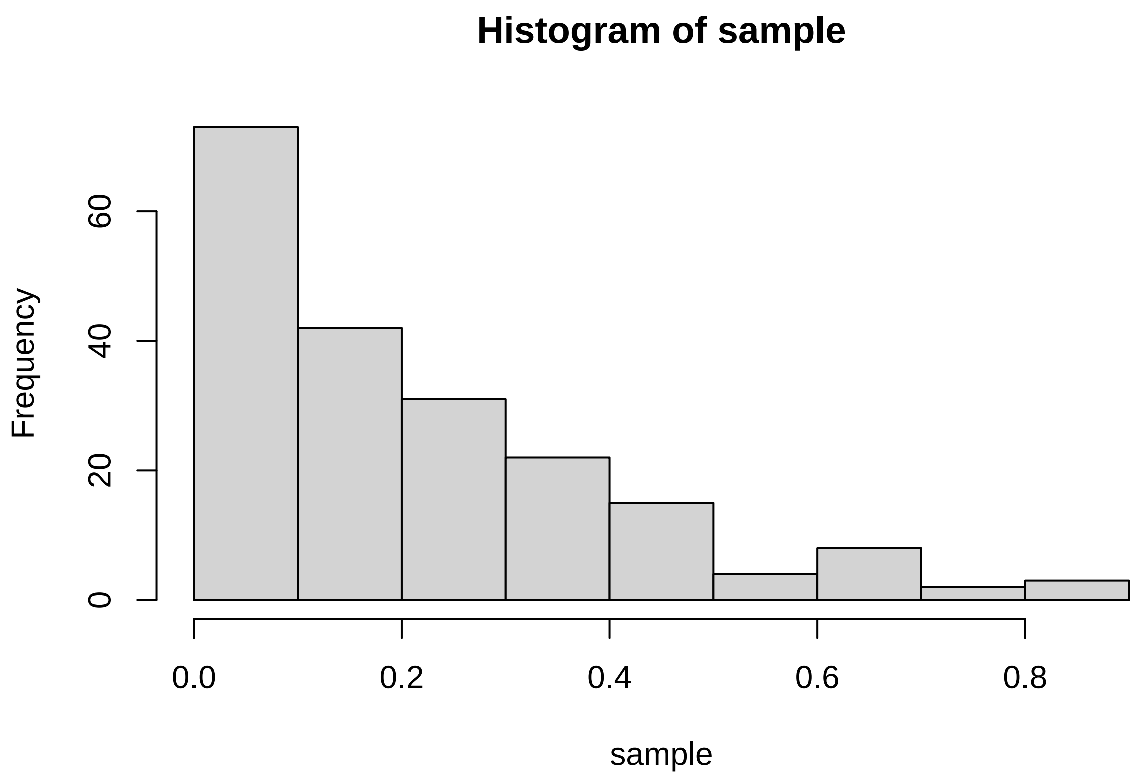
```
hist(X)
```



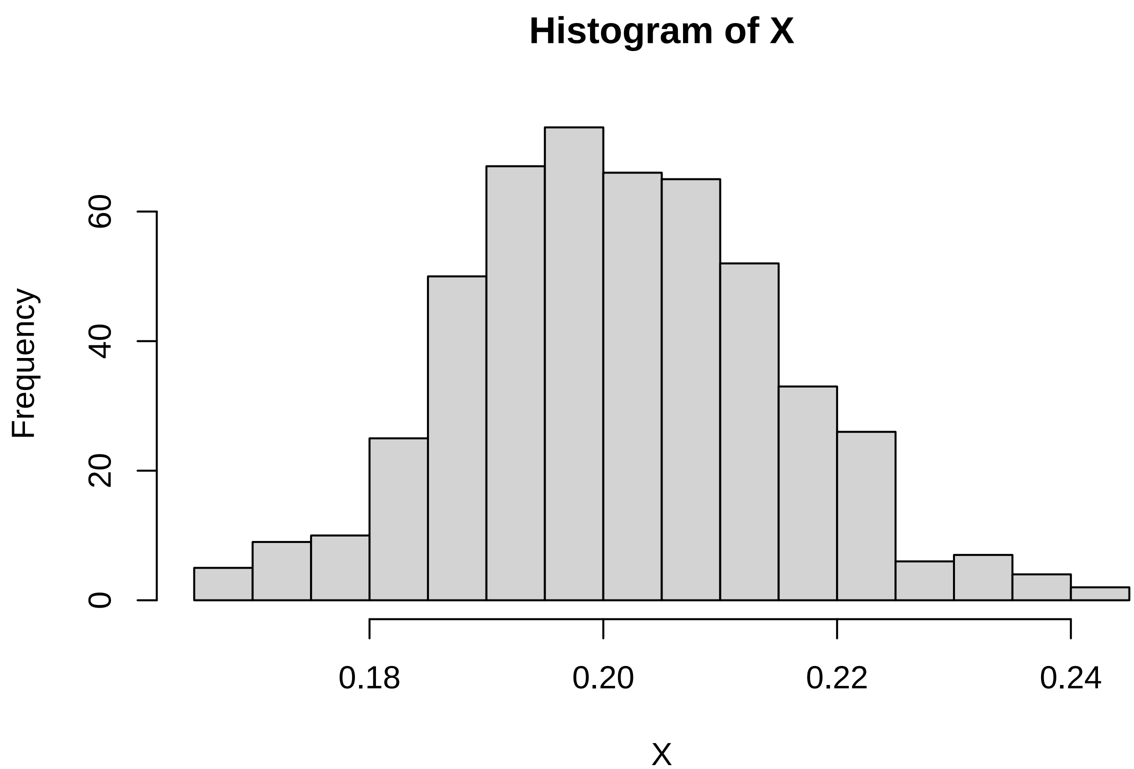
The data are still a bit right-skewed, but the shape is overall closer to a bell-shape than before.

4D

```
set.seed(10)
sample = rexp(n=200,rate=5) # Mostly the same, just for a larger sample size.
hist(sample)
```




```
R = 500
X = rep(NA,R) # This will store our Xbar values
for(i in 1:R)
{
  sample = rexp(n=200,rate=5)
  X[i] = mean(sample)
}
hist(X,breaks=20)
```



The population distribution is still right-skewed, but the estimator distribution is very close to a normal distribution, with little if any skew.